

Addendum: General Heterogeneity and the Power-Law Form of Bagged Excess Risk

Supplement to “Bagged Random Subset Forecast Combination under Common Loading”

13 Power-Law Approximation of Bagged Excess Risk under General Heterogeneity

13.1 Motivation

Section 7 derives the excess bagged risk approximation

$$R_k^{\text{bag},\infty} \approx C_\delta \left(\frac{1}{k} - \frac{1}{m} \right)^2 \quad (22)$$

under the small-heterogeneity assumption $\tau_i = \bar{\tau}(1 + \delta_i)$ with δ_i small. Simulations show that this approximation systematically underestimates the true excess risk when heterogeneity is large, and that the log-log regression slope of $R_k^{\text{bag},\infty}$ against $z_k = 1/k - 1/m$ is substantially below 2. This section provides a theoretical account of this discrepancy. We show that the true excess risk transitions between two asymptotic forms as heterogeneity grows, and that the power law

$$R_k^{\text{bag},\infty} \approx C \cdot z_k^\alpha, \quad z_k = \frac{1}{k} - \frac{1}{m}, \quad (23)$$

with estimated effective slope $\hat{\alpha}$ is a useful one-parameter summary of the excess risk curve across the practically relevant range of k .

13.2 Exact weight deviations

In the common-loading model, the exact bagged weight is

$$\bar{v}_{i,k} = \tau_i \cdot \frac{k}{m} \cdot \psi_i(k), \quad \psi_i(k) = E_{S \ni i, |S|=k} \left[\frac{1}{A_S} \right],$$

so the deviation from the full-pool optimum $w_i^{\text{opt}} = \tau_i/A_m$ is

$$\Delta_{i,k} := \bar{v}_{i,k} - w_i^{\text{opt}} = \tau_i \left(\frac{k}{m} \psi_i(k) - \frac{1}{A_m} \right). \quad (24)$$

The excess bagged risk is $R_k^{\text{bag},\infty} = \sum_i \Delta_{i,k}^2 / \tau_i$. The boundary conditions $\Delta_{i,m} = 0$ and $\Delta_{i,1} = \tau_i(1/(m\tau_i) - 1/A_m) = 1/m - w_i^{\text{opt}}$ hold exactly for every precision sequence $\{\tau_i\}$.

13.3 Two asymptotic regimes

Proposition 1 (Small-heterogeneity regime). *Under $\tau_i = \bar{\tau}(1 + \delta_i)$ with $\max_i |\delta_i|$ small,*

$$\Delta_{i,k} = -\frac{\delta_i}{m-1} z_k + O(\delta^2),$$

and consequently

$$R_k^{\text{bag},\infty} \approx C_\delta z_k^2, \quad C_\delta = \frac{BV_\delta}{m} \left(\frac{m}{m-1} \right)^2,$$

where $B = 1/\bar{\tau}$ and $V_\delta = m^{-1} \sum_i \delta_i^2$. The OLS log-log slope is $\hat{\alpha} = 2$.

Proof. This is equation (8) and equation (9) of the main text. $\square$

Proposition 2 (Large-heterogeneity regime). *Consider the dominant-forecaster model: $\tau_1 = \tau_H$ and $\tau_j = \tau_L$ for $j = 2, \dots, m$, with $\tau_H/\tau_L \rightarrow \infty$. Then for each $k \in \{1, \dots, m\}$:*

$$\Delta_{1,k} \approx -\frac{m-k}{m}, \quad \Delta_{j,k} \approx \frac{m-k}{m(m-1)}, \quad j \neq 1,$$

and

$$R_k^{\text{bag},\infty} \approx \frac{1}{(m-1)\tau_L} \left(1 - \frac{k}{m} \right)^2 = C' \left(1 - \frac{k}{m} \right)^2. \quad (25)$$

Proof. Forecaster 1 enters a random size- k subset with probability k/m . When included, its subset weight is $\tau_H/(\tau_H + (k-1)\tau_L) \rightarrow 1$ as $\tau_H/\tau_L \rightarrow \infty$; when excluded it receives weight zero. Hence $\bar{v}_{1,k} = (k/m) \cdot 1 + (1-k/m) \cdot 0 = k/m$, so $\Delta_{1,k} = k/m - w_1^{\text{opt}} \rightarrow k/m - 1 = -(m-k)/m$. For $j \neq 1$: forecaster j receives weight $\approx 1/k$ only in subsets that contain j but not forecaster 1. The unconditional probability of this event is

$$P(j \in S, 1 \notin S) = P(j \in S) \cdot P(1 \notin S \mid j \in S) = \frac{k}{m} \cdot \frac{m-k}{m-1}.$$

Hence

$$\bar{v}_{j,k} \approx \frac{k}{m} \cdot \frac{m-k}{m-1} \cdot \frac{1}{k} = \frac{m-k}{m(m-1)},$$

and $\Delta_{j,k} \approx (m-k)/(m(m-1))$ since $w_j^{\text{opt}} \rightarrow 0$. Inserting into $R = \Delta_{1,k}^2/\tau_H + (m-1)\Delta_{j,k}^2/\tau_L$ and letting $\tau_H \rightarrow \infty$ yields (25). $\square$

The two forms differ critically in how k enters. Writing $z_k = (m-k)/(km)$:

$$\text{Small het: } R_k \propto z_k^2; \quad \text{Large het: } R_k \propto k^2 z_k^2.$$

The large-heterogeneity form carries an extra factor k^2 relative to the small-heterogeneity form. For small k (large z), this factor is large, making R decay *much more slowly* than z^2 as k decreases from m .

13.4 Log-log slope of the excess risk: an OLS calculation

The empirical power-law $\hat{\alpha}$ comes from the OLS regression of $\log R_k$ on $\log z_k$ over $k \in \{2, \dots, m-1\}$. Define $u_k = \log(m-k)$ and $v_k = -\log k$, so that $\log z_k = u_k + v_k - \log m$ (since $z_k = (m-k)/(km)$).

Proposition 3 (OLS slope in the two limiting regimes). *Let $\bar{u}$, $\bar{v}$, V_u , V_v , C_{uv} denote the sample mean, variance, and covariance of (u_k, v_k) over $k \in \{2, \dots, m-1\}$.*

- (i) *If $\log R_k = \text{const} + 2 \log z_k$ (small-het), then $\hat{\alpha} = 2$ exactly.*
- (ii) *If $\log R_k = \text{const} + 2u_k$ (large-het, since $R \propto (m-k)^2$), then*

$$\hat{\alpha} = \frac{2(V_u + C_{uv})}{V_u + 2C_{uv} + V_v}.$$

When $V_u = V_v$ (which holds approximately by symmetry of the range), $\hat{\alpha} = 1$ exactly.

Proof. (i) Immediate since $\log R = c + 2 \log z$ is linear in $\log z$ with slope 2.

(ii) $\log R = c + 2u$ and $\log z = u + v - \log m$, so

$$\hat{\alpha} = \frac{\text{Cov}(2u, u+v)}{\text{Var}(u+v)} = \frac{2(V_u + C_{uv})}{V_u + 2C_{uv} + V_v}.$$

Since $u_k = \log(m-k)$ and $v_k = -\log k$ both decrease in k over the symmetric range $k \in \{2, \dots, m-1\}$, we have $C_{uv} > 0$. Setting $V_u = V_v \equiv V$ gives $\hat{\alpha} = 2(V + C_{uv})/(2V + 2C_{uv}) = 1$. $\square$

Remark 1. *The symmetry $V_u \approx V_v$ holds for any m because the distribution of $\log(m-k)$ over $k = 2, \dots, m-1$ is the reversal of the distribution of $\log k$ over the same range; they have the same variance. Hence $\hat{\alpha} \rightarrow 1$ as $\tau_H/\tau_L \rightarrow \infty$, regardless of m .*

13.5 Summary: the power law as an interpolation

The three regimes are:

Regime	True form of R_k	Eff. log-log slope $\hat{\alpha}$	k^* scaling [†]
Homoskedastic	0	—	any k optimal
Small het	$C_\delta z_k^2$	2 (exact)	$T^{1/3}$
Large het (dominant)	$C' \left(\frac{mz_k}{1+mz_k} \right)^2$	≈ 1 (grid OLS)	$T^{1/2}$

[†] Scaling of k^* under the *fitted* power law $R \approx Cz^{\hat{\alpha}}$; see text.

For intermediate heterogeneity, R_k is a weighted mixture of both forms. The weight deviation of a near-dominant forecaster ($\tau_i \gg \bar{\tau}$) contributes $O(k^2 z^2)$, while near-equal forecasters contribute $O(z^2)$. The two analytically tractable limits give effective OLS slopes of 2 (small het) and ≈ 1 (dominant forecaster, grid OLS); for intermediate heterogeneity, simulations consistently deliver $\hat{\alpha}$ between these values, though a general proof that $\hat{\alpha} \in [1, 2]$ for arbitrary $\{\tau_i\}$ remains open.

Remark 2 (What the dominant-forecaster curve actually is). *The exact expression $R_k = C'(1 - k/m)^2 = C'(kz_k)^2$ can be rewritten as $R_k = C'(mz_k/(1 + mz_k))^2$ since $(1 - k/m) = mz_k/(1 + mz_k)$. This is not a power law in z_k : locally near $k = m$ it behaves as $C'm^2z_k^2$ (slope 2), while for small k it saturates at approximately C' . The effective OLS slope $\hat{\alpha} \approx 1$ is therefore a finite-grid projection artefact, not a true asymptotic exponent. The power law $R \approx Cz^{\hat{\alpha}}$ should be understood as a practical summary of the excess risk curve, not as an exact functional form.*

The power law with estimated $\hat{\alpha}$ provides a practical one-parameter summary of the excess risk curve. Simulations with $m = 50$, $T = 240$, and heterogeneity ratio $H \in \{1.05, 6\}$ give $\hat{\alpha} \in \{1.05, 1.66\}$ with $R^2 > 0.98$ in all cases, confirming the adequacy of the approximation over the practically relevant range of k .

13.6 Optimal subset size under the power law

Minimizing the finite-sample bagged loss

$$L_k^{\text{bag}, \infty} = (A + Cz_k^\alpha) \frac{T-1}{T-k}, \quad A = s + Q^{\text{opt}},$$

with respect to k (treating k as continuous) and using $dz/dk = -1/k^2$, the interior first-order condition is:

$$C\alpha z^{\alpha-1} \cdot \frac{T-1}{k^2(T-k)} = (A + Cz^\alpha) \frac{T-1}{(T-k)^2}.$$

For $k \ll m$ (so $z \approx 1/k$) and $T \gg k$ (so $T - k \approx T$), the leading-order balance gives:

$$C\alpha \cdot \frac{T}{k^{\alpha+1}} \approx A \implies k^* \approx \left(\frac{C\alpha T}{A} \right)^{1/(\alpha+1)}. \quad (26)$$

This generalizes the small-heterogeneity cube-root rule (14) of the main text. The table below gives the implied T -scaling under the *fitted* power law $R \approx Cz^{\hat{\alpha}}$:

$\hat{\alpha}$	Setting	k^* scaling (under fitted power law)
2	Small het (eq. (14))	$T^{1/3}$
3/2	Intermediate	$T^{2/5}$
1	Large het (grid OLS)	$T^{1/2}$

Remark 3 (True dominant-forecaster optimum differs). *When $R_k = C'(1 - k/m)^2$ exactly (dominant-forecaster limit), minimizing $L_k = (A + C'(1 - k/m)^2)(T-1)/(T-k)$ directly gives $m - k^* \approx Am^2/(2C'T)$, so $k^* \rightarrow m$ at rate $1/T$. The square-root scaling $k^* \propto T^{1/2}$ in the table above is the prediction of the fitted power-law approximation $R \approx Cz$, not of the true dominant-forecaster loss; the two differ because the fitted power law misrepresents the saturation of R at small k . In practice one should use the fitted power-law formula as a summary tool over the observed range of k , not extrapolate its T -scaling to the extreme dominant-forecaster limit.*

Under the fitted power law, the cube-root rule $k^* \propto T^{1/3}$ is a *lower bound* on the T-scaling exponent. As heterogeneity grows and $\hat{\alpha}$ falls toward 1, k^* (under the fitted approximation) grows faster with T , toward the square-root rule of the average-subset estimator (Proposition 9.1).

Remark 4 (Practical estimation of α and k^*). *When the DGP is unknown, estimate α by:*

- (i) *Compute $R_k^{\text{bag},\infty}$ by Monte Carlo for a sparse grid $k \in \{2, 5, 10, \lceil m/3 \rceil, \lceil m/2 \rceil\}$.*
- (ii) *Run OLS of $\log R_k^{\text{bag},\infty}$ on $\log z_k$ to obtain $\hat{\alpha}$ and $\hat{C}$.*
- (iii) *Compute $k^* = \lfloor (\hat{C}\hat{\alpha}T/A)^{1/(\hat{\alpha}+1)} \rfloor$ and verify by evaluating L_k at $k^* \pm 2$.*

In the baseline exogenous simulations ($m = 50$, $T = 240$), this procedure recovers $k^ = 3$ (low heterogeneity, $H = 6$, $B = 3$) and $k^* = 6$ (high heterogeneity, $H = 6$, $B = 8$), matching the exact grid minimizer in both cases.*

Remark 5 (Connection to the finite-bagging optimum). *For finite G bags, Proposition 1 of the main text gives $Q_k^{\text{bag},G} = Q_k^{\text{bag},\infty} + G^{-1}(Q_k^{\text{sub}} - Q_k^{\text{bag},\infty})$. Using $Q_k^{\text{sub}} - Q_k^{\text{opt}} \approx B(1/k - 1/m)$ (average-subset leading term) and $R_k^{\text{bag},\infty} \approx Cz^\alpha$:*

$$R_k^{\text{bag},G} \approx \frac{B}{G} z_k + \left(1 - \frac{1}{G}\right) C z_k^\alpha.$$

This nests the general-heterogeneity finite-bagging expression (eq. (10)) of the main text when $\alpha = 2$. For finite G , the first-order (z) finite-bagging term dominates for small k , and the optimal k for finite bagging transitions from the square-root rule (small G) to the power-law rule (large G), as in Section 9.3 of the main text.